

1. BACKGROUND

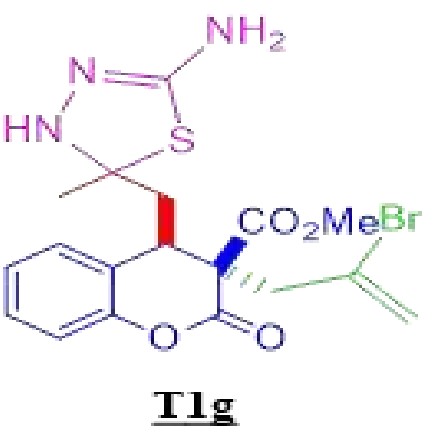
Alzheimer's disease is associated with progressive impairment of cholinergic neurotransmission. Acetylcholinesterase (AChE) inhibition remains an established strategy for increasing acetylcholine availability.

Coumarin–1,3,4-thiadiazole hybrids provide an attractive scaffold for the exploration of new AChE inhibitors.

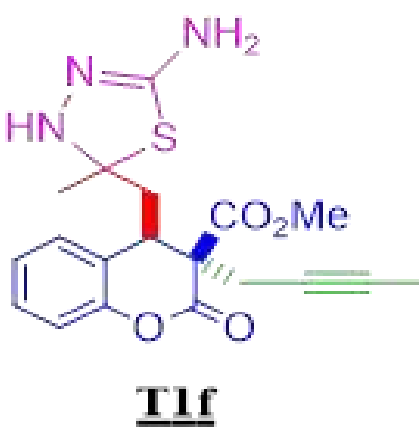
2. AIM OF THE STUDY

To evaluate the AChE inhibitory activity of novel coumarin–1,3,4-thiadiazole derivatives and to investigate the molecular interactions and predicted ADMET properties of the most promising compounds.

3. SELECTED COMPOUNDS



AChE IC₅₀ = 20.35 ± 0.06 μM



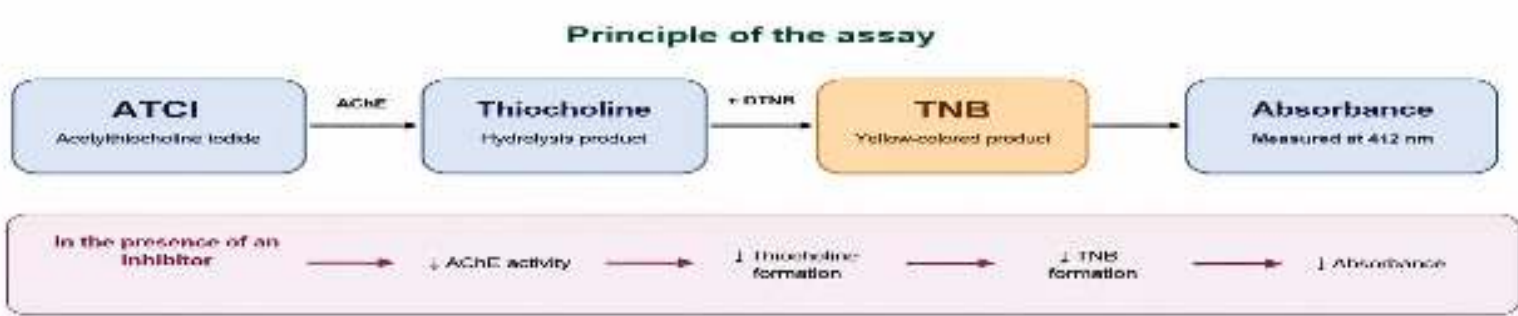
AChE IC₅₀ = 38.90 ± 0.46 μM

T1g and T1f were the most active synthesized derivatives against AChE. T1g showed the strongest inhibitory activity of the series, followed by T1f.

Galantamine (reference): IC₅₀ = 4.14 ± 0.07 μM.

4. ELLMAN'S AChE INHIBITION ASSAY

AChE hydrolyzes acetylthiocholine iodide (ATCI) to produce thiocholine. Thiocholine reacts with DTNB to form the yellow TNB product, measured at 412 nm. Inhibition decreases TNB formation and consequently reduces absorbance.



Experimental ranking: T1g (20.35 μM) > T1f (38.90 μM) > T3a (86.33 μM) > T2c (121.72 μM) > T3b (143.71 μM).

5. KEY FINDINGS

T1g	Most active synthesized derivative	IC ₅₀ = 20.35 ± 0.06 μM
T1f	Second most active derivative	IC ₅₀ = 38.90 ± 0.46 μM
Docking	Favorable predicted binding	T1g showed more favorable scores
ADMET	Favorable drug-likeness profile	High GI absorption; 0 Lipinski violations

6. CONCLUSION

T1g and T1f were identified as the most active synthesized derivatives against AChE, with T1g showing the strongest inhibitory activity.

Molecular docking supported the favorable binding potential of both compounds, with T1g exhibiting consistently more favorable predicted docking scores than T1f.

ADMET predictions indicated favorable drug-likeness, high gastrointestinal absorption and no predicted hepatotoxicity. However, the predicted lack of BBB permeation and P-gp substrate liability remain important considerations for CNS-oriented optimization.

Overall, T1g represents the most promising lead of the investigated series for further structural optimization and experimental validation as an AChE inhibitor.

7. MOLECULAR DOCKING — T1g & T1f

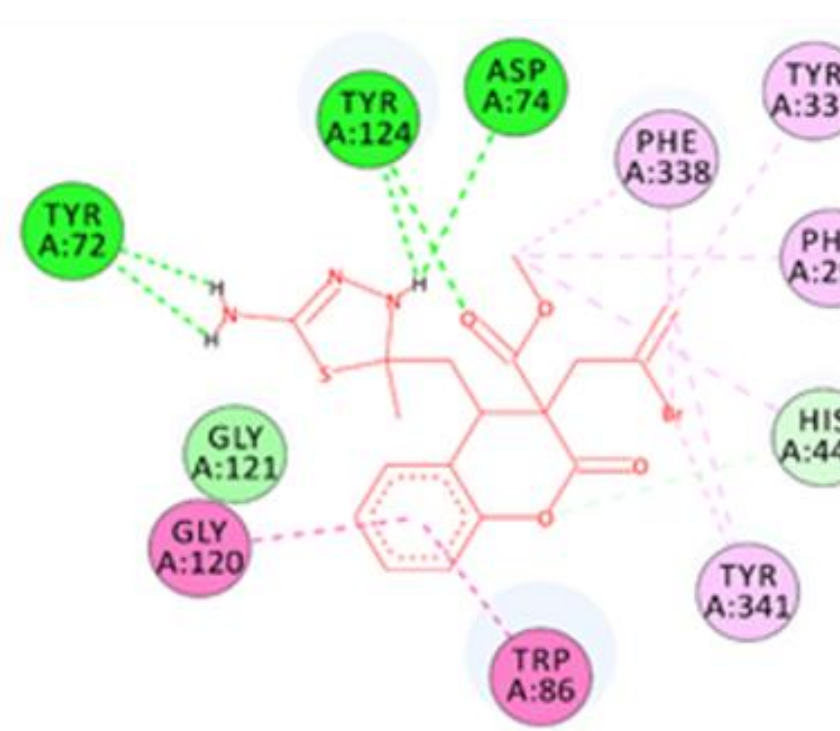
AutoDockTools 1.5.6 • Protein targets: 4EY7, 1P0I and 1DNU

Compound	4EY7	1P0I	1DNU
T1g	-12.1	-9.7	-7.7
T1f	-10.1	-8.3	-7.0

T1g showed more favorable predicted docking scores than T1f across all three investigated targets.

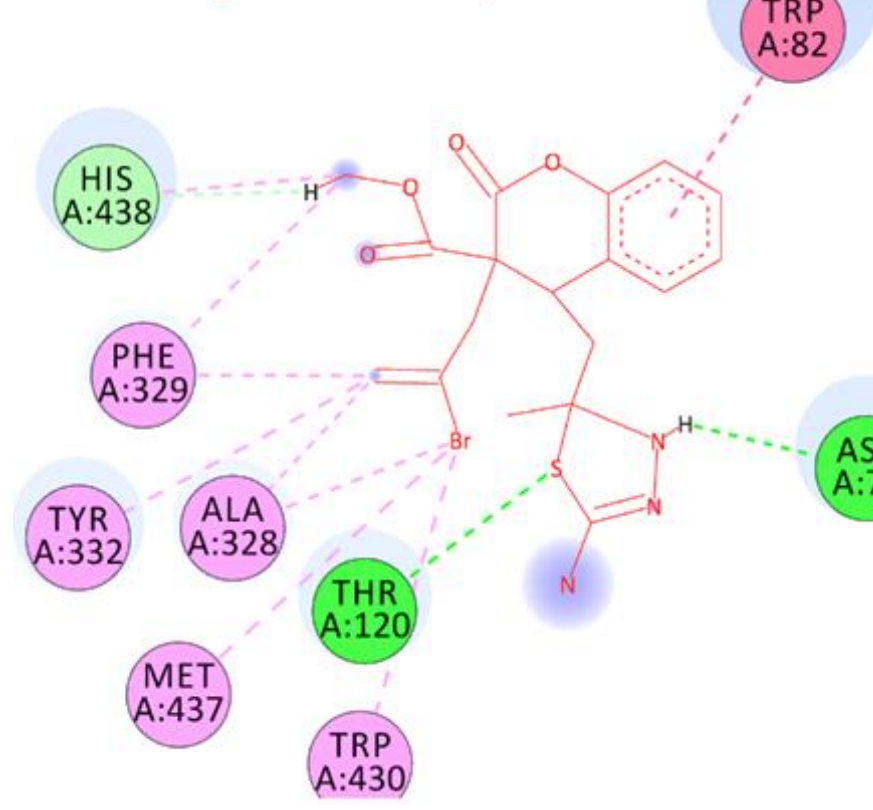
Ligand–Protein Interaction Diagrams

4EY7

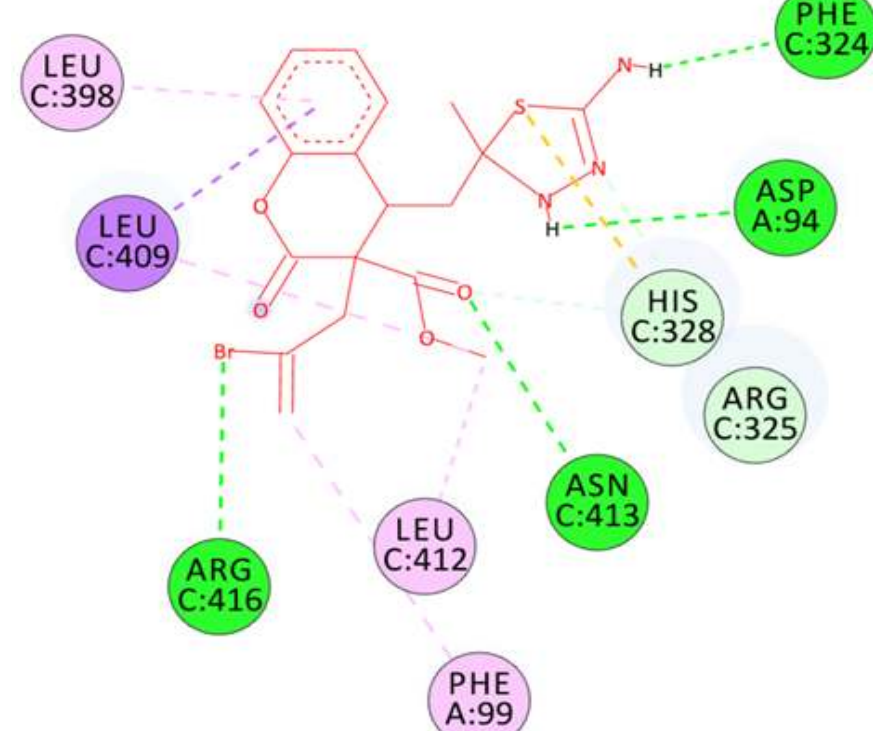


T1g
IC₅₀ =
20.35 ±
0.06 μM

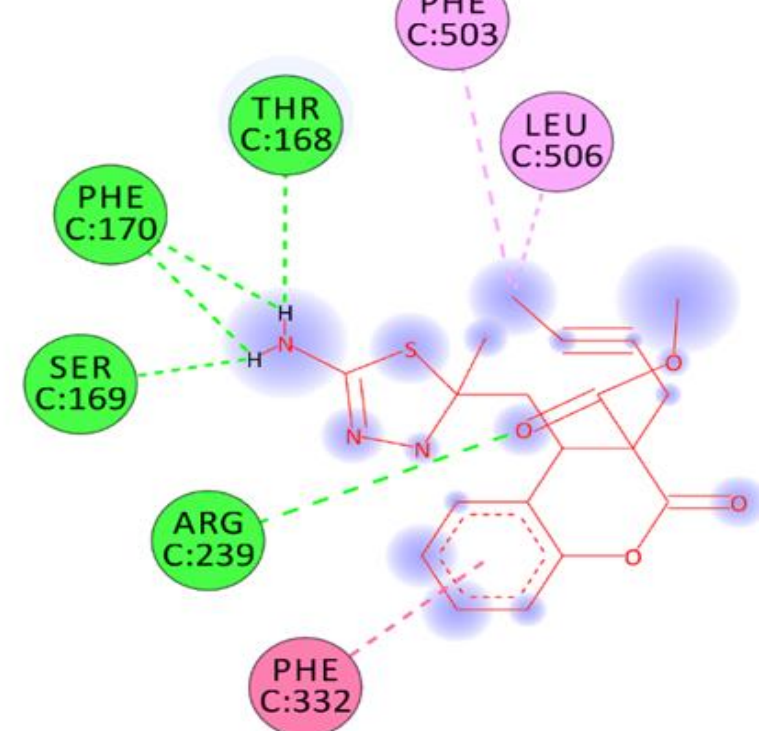
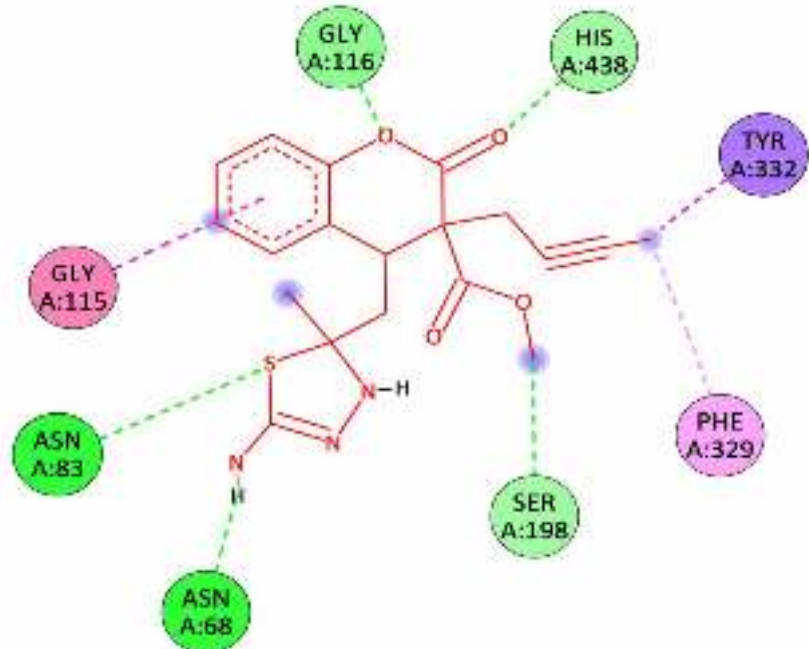
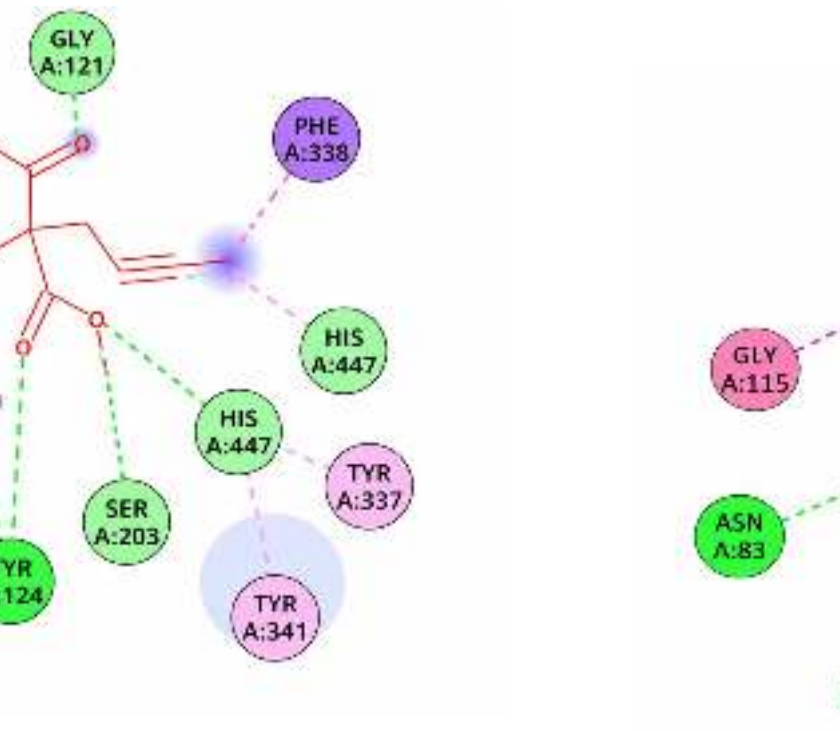
1P0I



1DNU



T1f
IC₅₀ =
38.90 ±
0.46 μM



8. IN SILICO ADMET — T1g & T1f

SwissADME and ProTox-II

Parameter	T1g	T1f
TPSA (Å²)	128.31	128.31
MLogP	2.62	2.50
GI absorption	High	High
BBB permeation	No	No
P-gp substrate	Yes	Yes
CYP1A2 inhibitor	No	No
CYP3A4 inhibitor	Yes	Yes
Lipinski violations	0	0
Hepatotoxicity	Inactive	Inactive

Favorable: high GI absorption, 0 Lipinski violations and no predicted hepatotoxicity.
Considerations: no predicted BBB permeation, P-gp substrate liability and predicted CYP3A4 inhibition.

9. MAIN MESSAGE

T1g emerged as the lead compound of the series.

IC₅₀ = 20.35 ± 0.06 μM

More favorable docking scores than T1f
+ multiple predicted ligand–protein interactions
+ favorable drug-likeness profile

T1g is therefore the priority compound for further structural optimization and experimental validation.